A.I. and I.T

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Overview

Training A.I. Models

Building A.I. Apps

Using A.I. Tools

Training A.I. Models

Training A.I. Models

Training via data set

Training via simulation

Training via demonstration

jupyter Training an AI Model via Data Set Last Checkpoint: 9 minutes ago (autosaved)

Logout

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3

Run Code

Training an A.I. Model Using a Data Set

1. Load the Data

In [1]:
```
import os
import pandas as pd
import seaborn as sns
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.model_selection import GridSearchCV
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
```

In [3]:
```
os.chdir("C:\Workshop\Data")
```

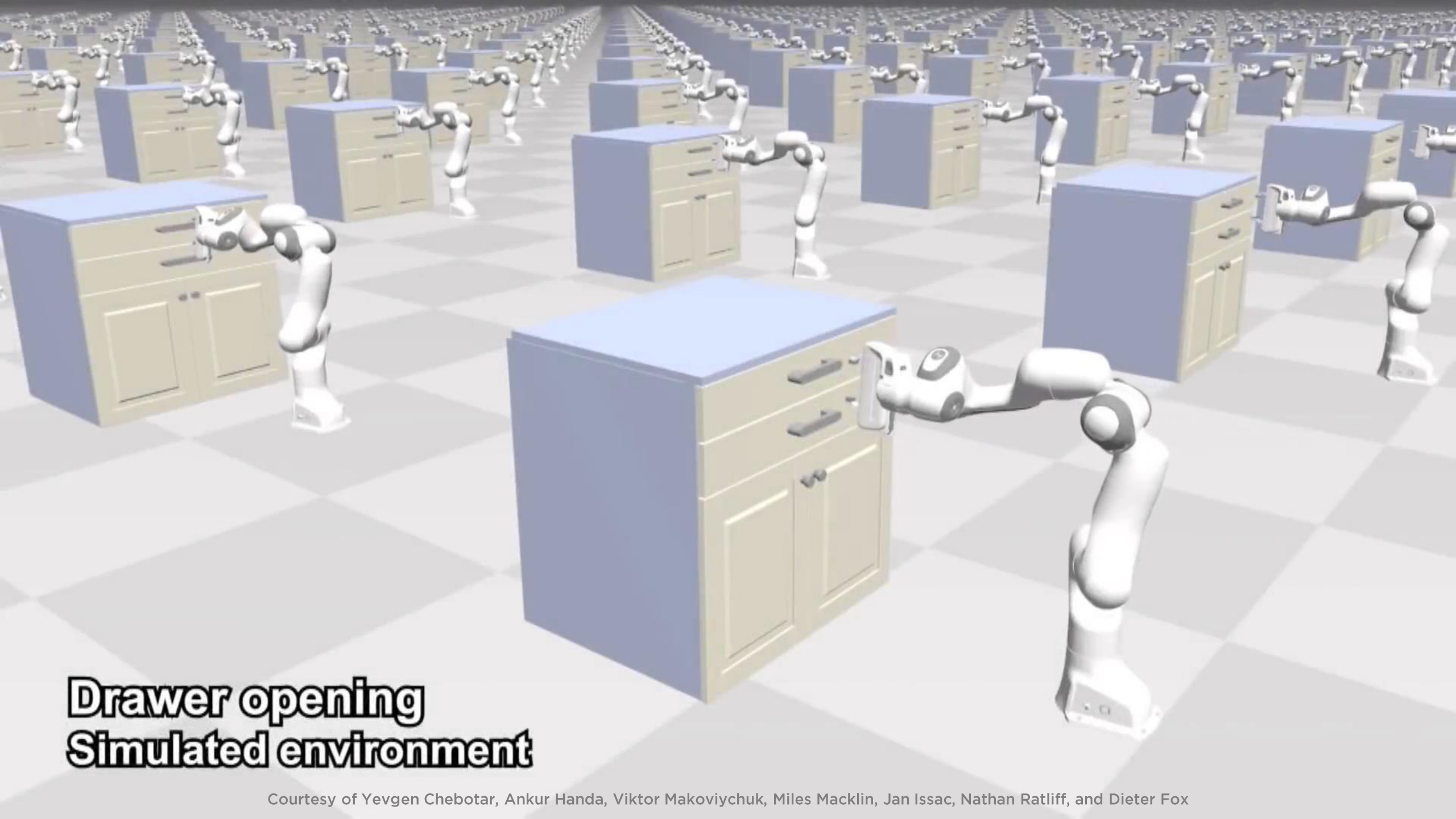
In [4]:
```
policies = pd.read_csv("Risk.csv")
```

2. Transform the Data

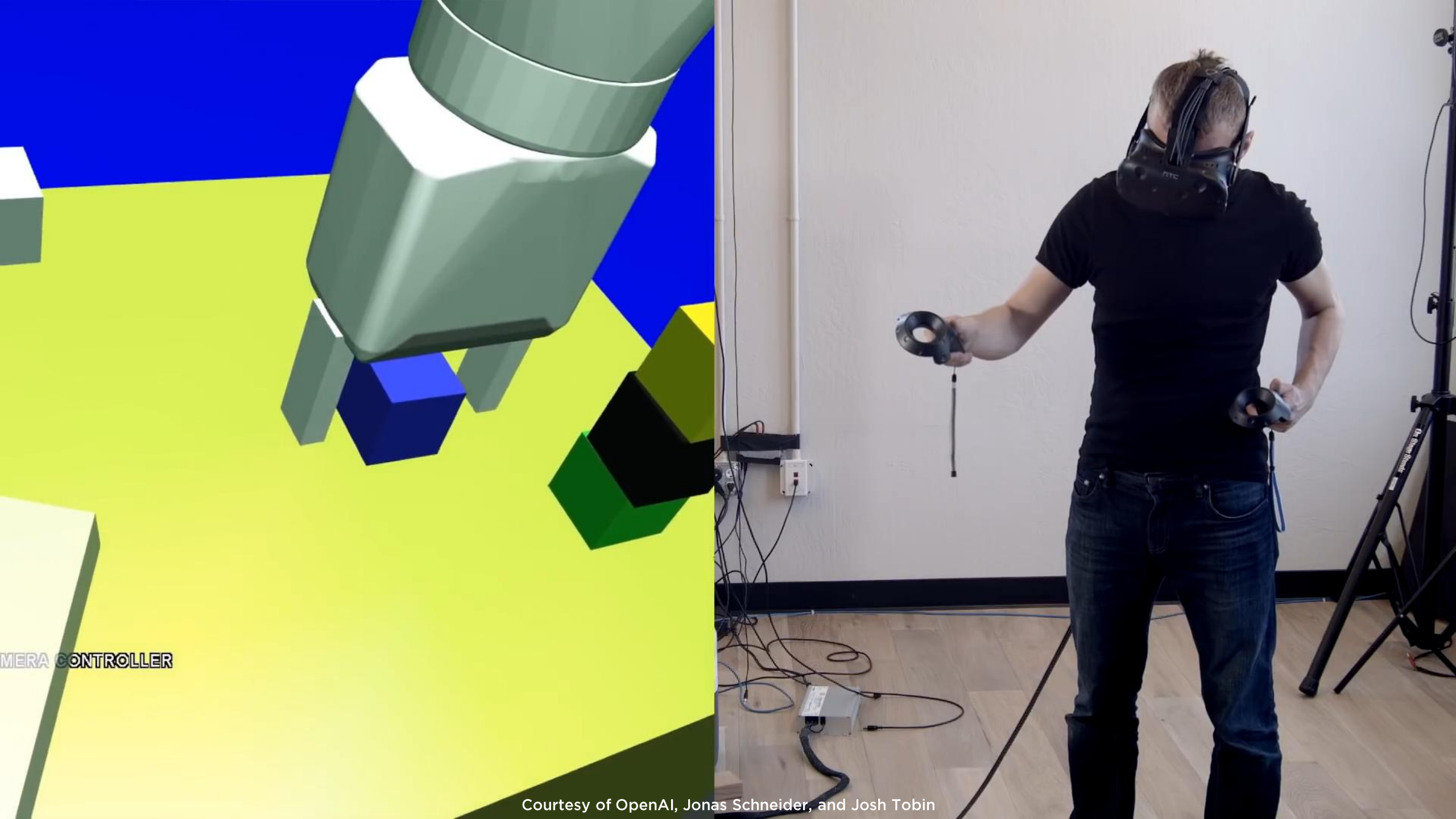
In [5]:
```
X = policies[["Gender", "State_Rate", "Height", "Weight", "BMI", "Age"]]
```

In [6]:
```
X.Gender = X.Gender.replace(("Female", "Male"), (0, 1))
```

In [7]:
```
y = policies.Risk
```



Courtesy of Yevgen Chebotar, Ankur Handa, Viktor Makoviychuk, Miles Macklin, Jan Issac, Nathan Ratliff, and Dieter Fox



Courtesy of OpenAI, Jonas Schneider, and Josh Tobin

Courtesy of the HAHN Group

Training A.I. Models

Training via data set

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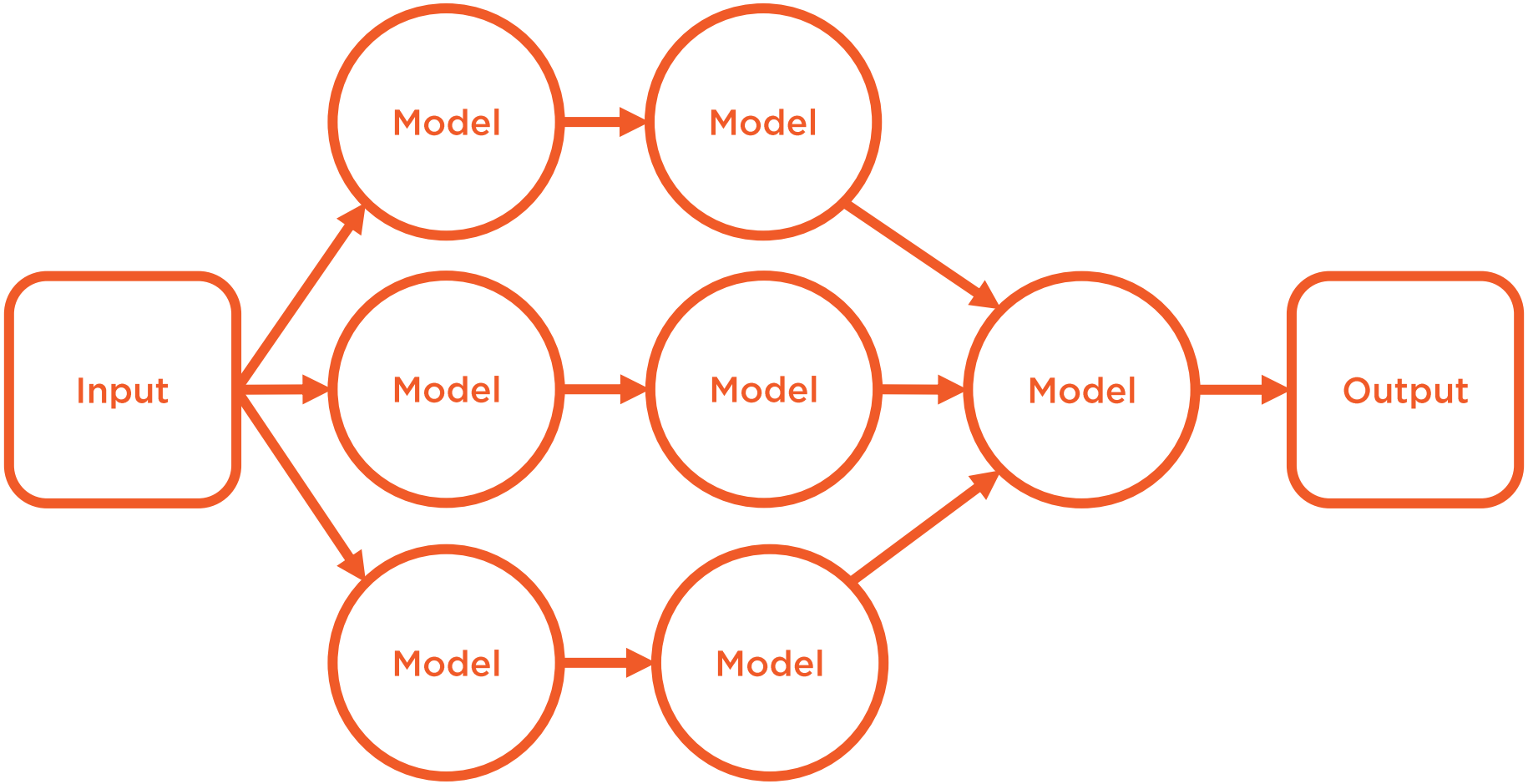
Building A.I. Apps

Modular A.I. Applications



- ☑ Image contains product
- ☑ Customers are happy
- ☑ Sentiment is positive

Big Cola, Inc

Thanks for sharing your big day with us, John! :)

Here's a free big cola for your next party!

http://bigcola.com/redeem

Post scanner

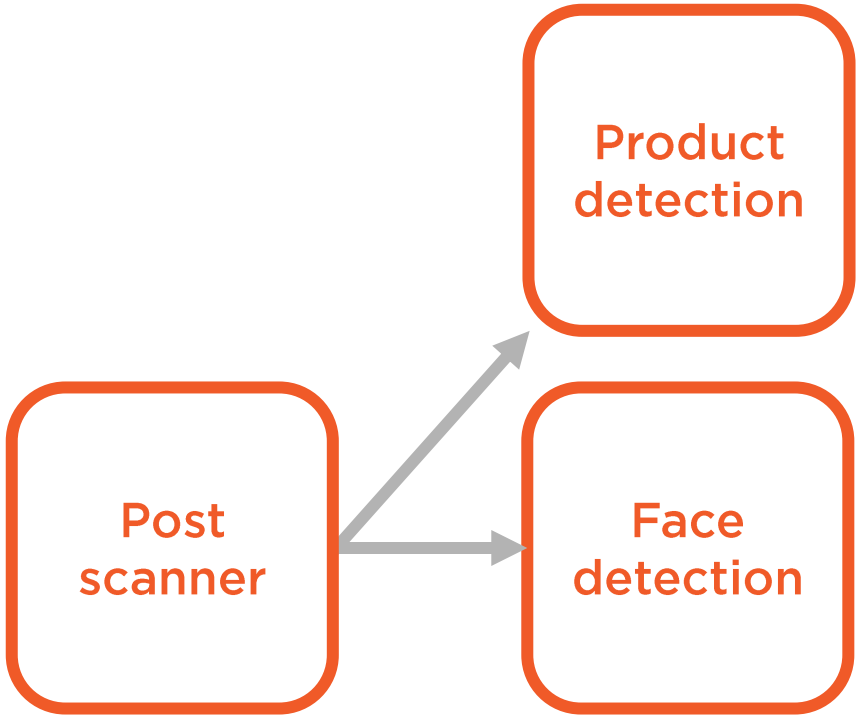
My friends and I celebrated my 21st birthday
by sharing a #BigCola and playing "cola pong"!

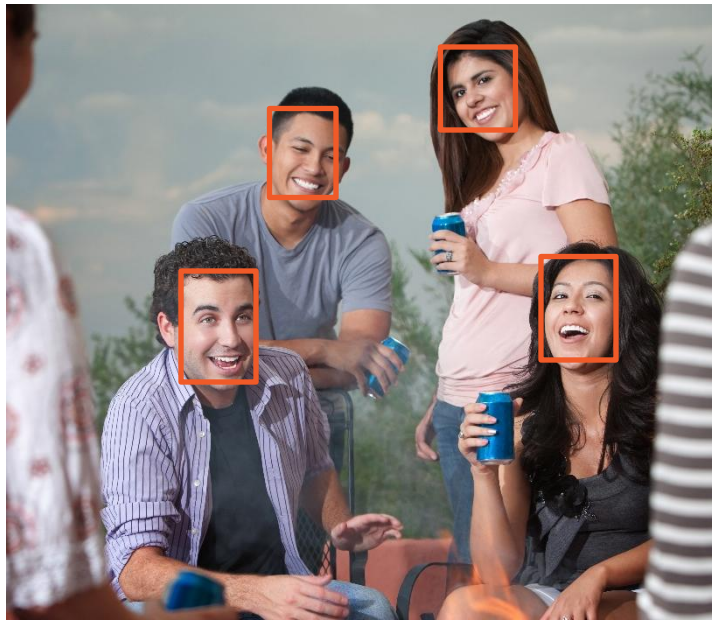
Brand detected
(100% confidence)

Post scanner

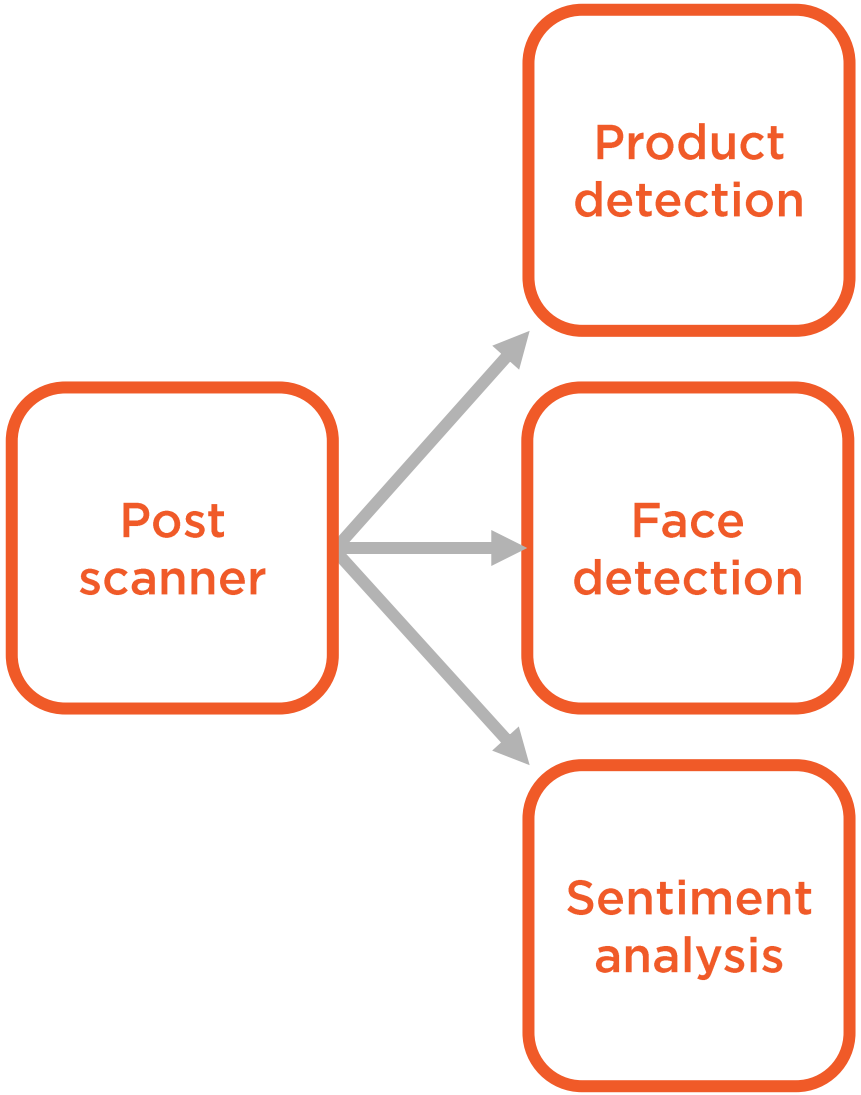
Product detection

Product detected (90% confidence)



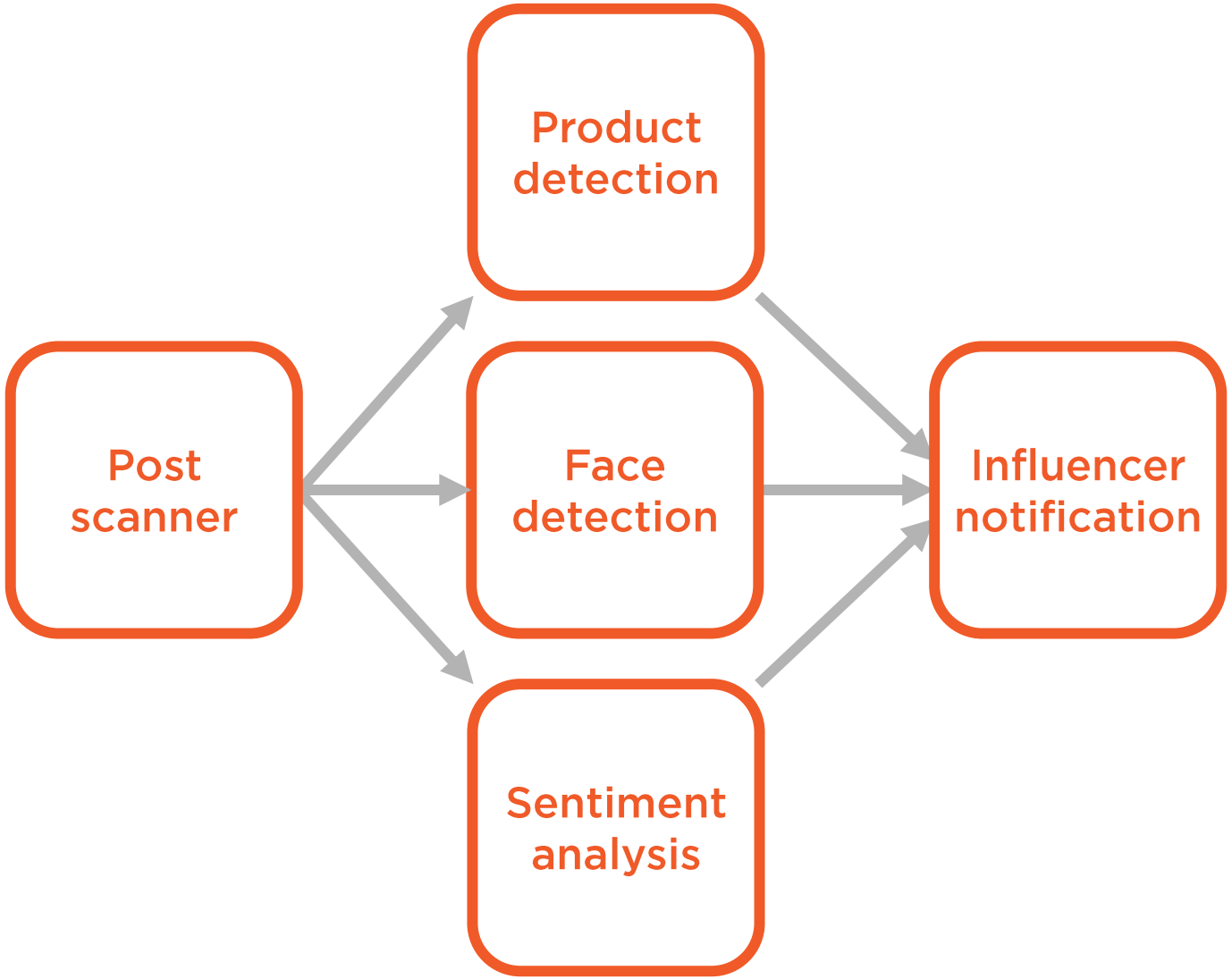


Emotion: happiness
(99% confidence)



My friends and I celebrated my 21st birthday
by sharing a #BigCola and playing "cola pong"!

Sentiment: positive
(95% confidence)



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http://bigcola.com/redeem

It’s the combination of A.I. modules that will lead to A.I. applications of the future.

Pre-trained A.I. Modules

Object detection

Face recognition

Emotion detection

Gesture recognition

Pose estimation

Video indexing

Text analysis

Language translation

Text to speech

Speech to text

Speaker recognition

Handwriting recognition

And more ...

A.I. Model Repositories

Hugging Face

PyTorch Hub

ONNX Model Zoo

Hosted A.I. Services

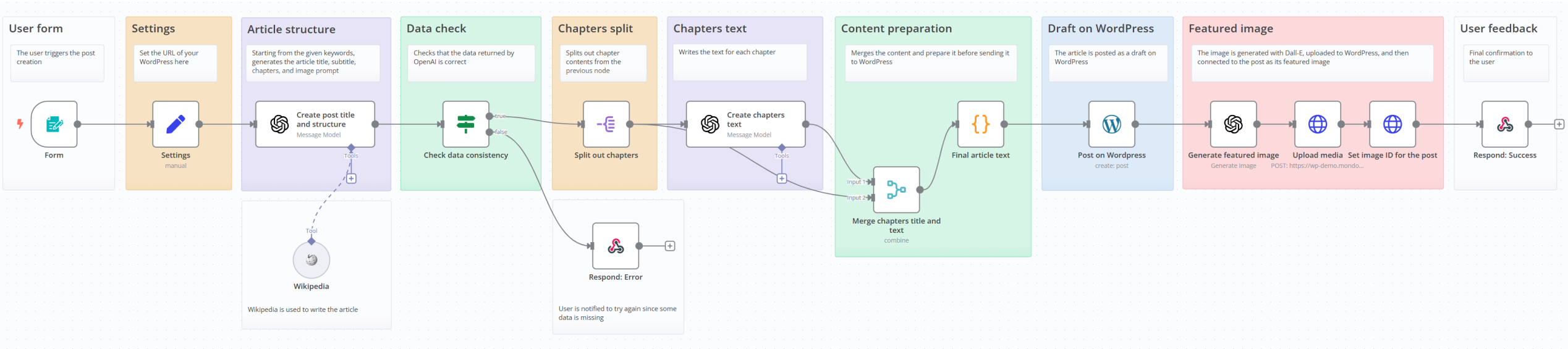
Microsoft
Azure AI
Services

Google
Cloud AI
Services

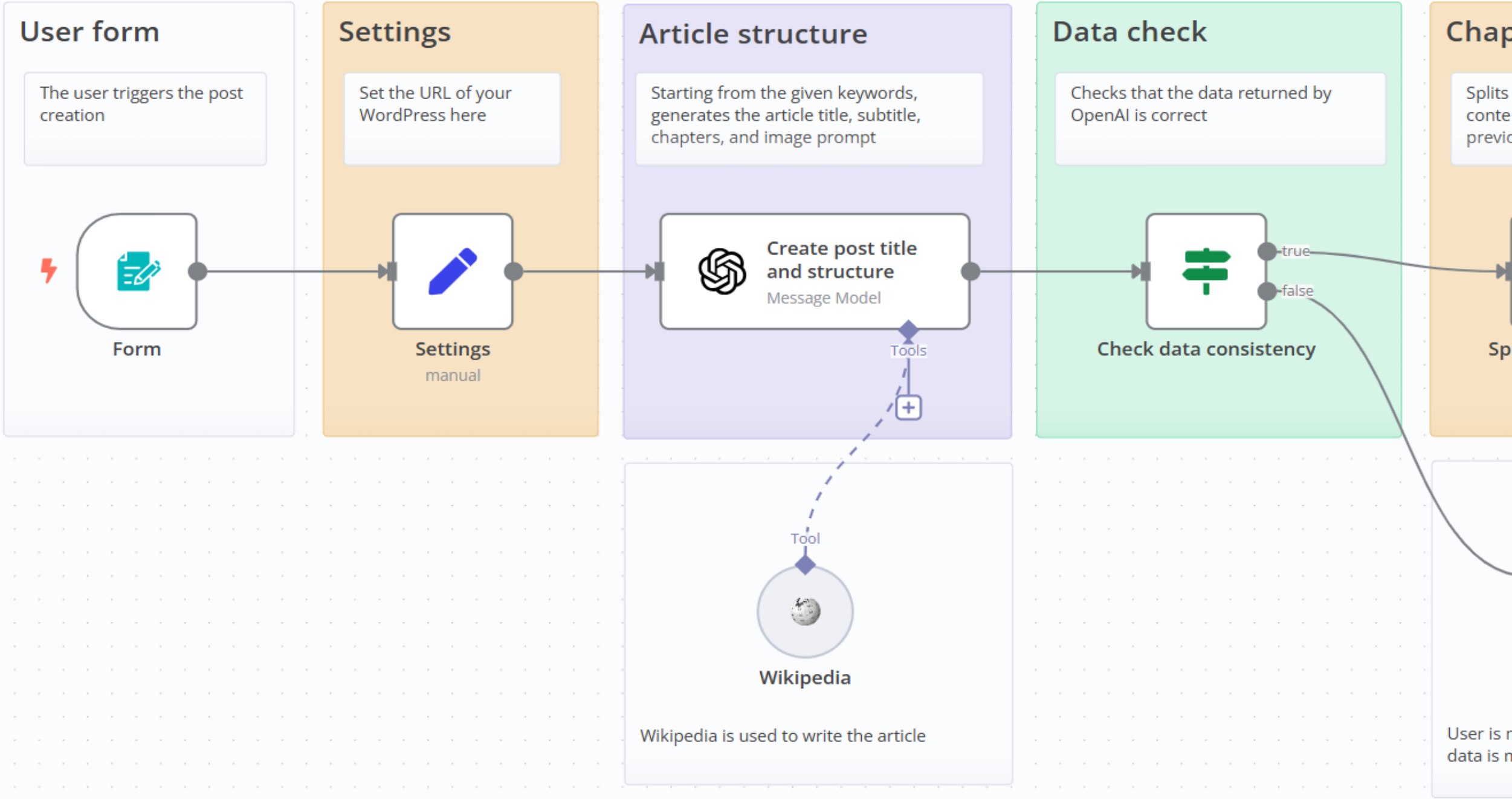
Amazon
AWS AI
Services

Generate a WordPress article from a list of keywords.

Source: n8n


User form
The user triggers the post creation
Form
Settings
Set the URL of your WordPress here
Settings
manual
Article structure
Starting from the given keywords, generates the article title, subtitle, chapters, and image prompt
Create post title and structure
Message Model
Tools
Tool
Wikipedia
Wikipedia is used to write the article
Data check
Checks that the data returned by OpenAI is correct
Check data consistency
true
false
Chapters split
Splits out chapters contents from the previous node
Split out chapters
Respond: Error
User is notified to try again since some data is missing
Chapters text
Writes the text for each chapter
Create chapters text
Message Model
Tools
Content preparation
Merges the content and prepare it before sending it to WordPress
Input 1
Input 2
Merge chapters title and text
combine
Final article text
Draft on WordPress
The article is posted as a draft on WordPress
Post on Wordpress
create: post
Featured image
The image is generated with Dall-E, uploaded to WordPress, and then connected to the post as its featured image
Generate featured image
Generate Image
Upload media
POST: https://wp-demo.mondo...
Set image ID for the post
User feedback
Final confirmation to the user
Respond: Success

Source: n8n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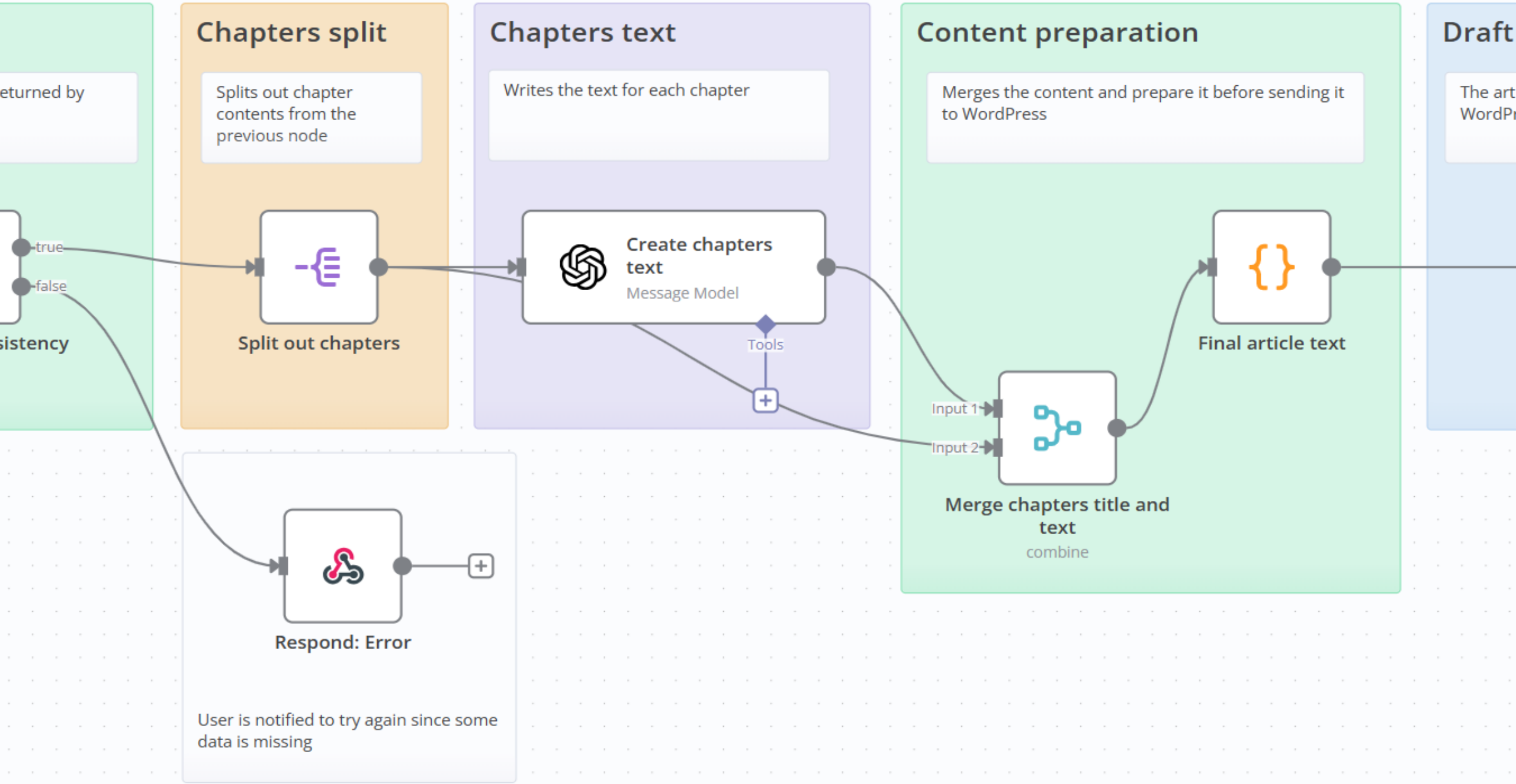
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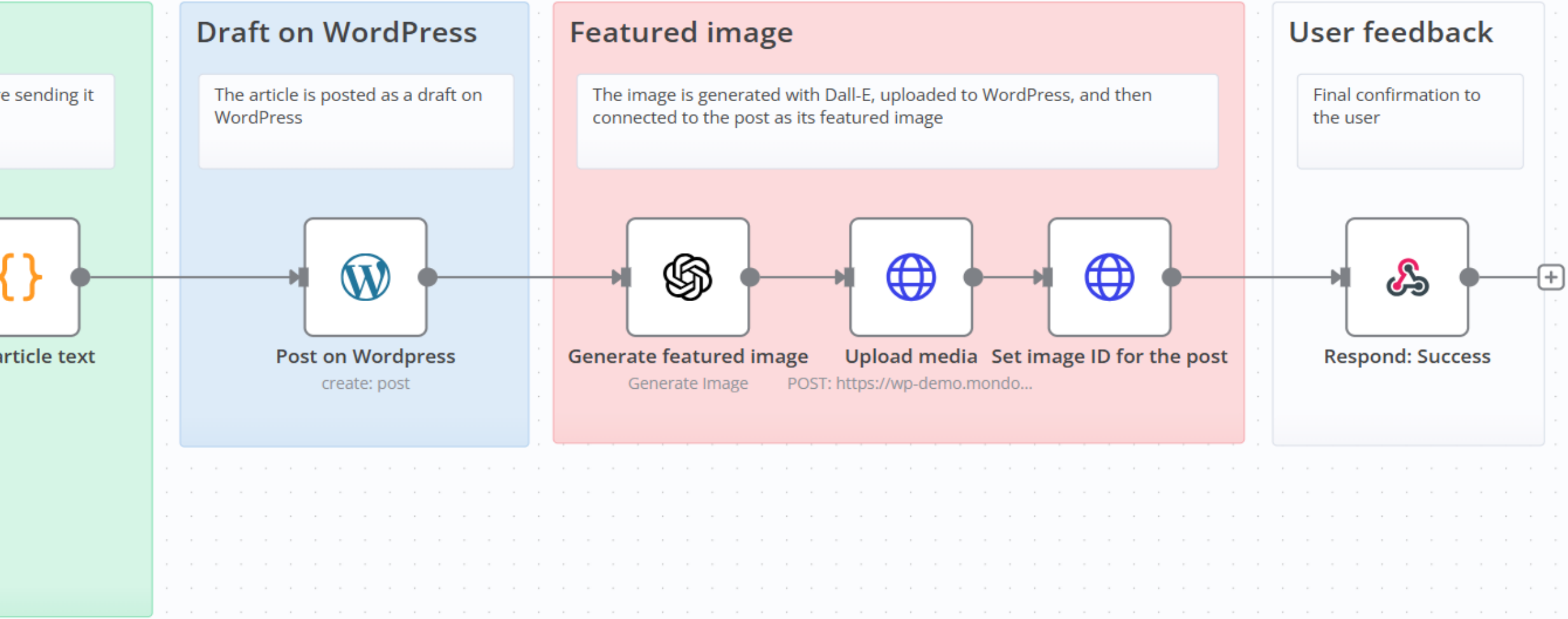
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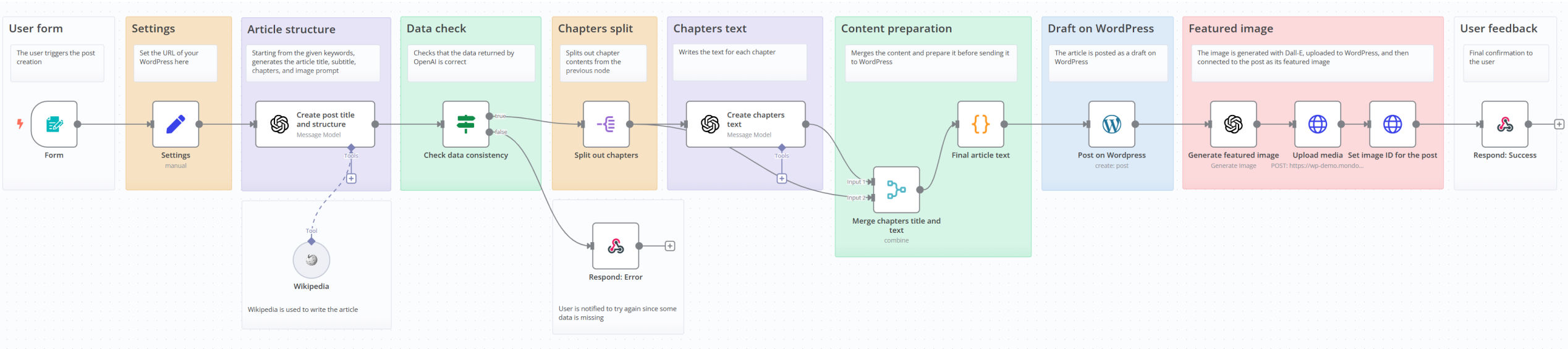
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Source: n8n

Everything you need to get started
is available today!

Using A.I. Tools

If you can’t beat ‘em… join ‘em.

Don’t try competing *against* A.I. ...

... learn to work *with* A.I. instead.

ChatGPT 4o

Share

PLUS

What is AI?

Artificial Intelligence (AI) is the field of computer science focused on creating systems that can perform tasks that typically require human intelligence. These tasks include:

- **Understanding language** (natural language processing)
- **Recognizing patterns** (e.g., in images or audio)
- **Making decisions** (planning, reasoning, and problem-solving)
- **Learning from experience** (machine learning)
- **Interacting with environments** (robotics and agents)

In simple terms, AI is about building machines that can think, learn, and act in ways that mimic or extend human capabilities.

Ask anything

Tools

ChatGPT can make mistakes. Check important info.

```python
import os
import sys

# Count lines of code in the given directory, separated by file extension
def main(directory):
    line_count = {}
```

Source: TabNine

Avengers - Power Query Editor

File | Home | Transform | Add Column | View

Column From Examples | Custom Column | Invoke Custom Function | Conditional Column | Index Column | Duplicate Column
General

Format | Merge Columns | Extract | Parse
From Text

Statistics | Standard | Scientific | Trigonometry | Rounding | Information
From Number

Date | Time | Duration
From Date & Time

Queries

	Super Name	Real Name
1	Captain America	Capt. Steve Rogers
2	Iron Man	tOnY sTaRk
3	The Hulk	Dr. Bruce Banner
4	Thor	THOR OF ASGARD
5	Black Widow	natasha romanoff
6	Hawkeye	Mr. Clint Barton
7	Dr. Strange	Dr. Steven Strange
8	Spiderman	PETER parker
9	Black Panther	T'Challa
10	Captain Marvel	Capt. Carol Danvers

Query Settings

PROPERTIES

Name

Avengers

All Properties

APPLIED STEPS

Source

Changed Type

2 COLUMNS, 10 ROWS

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Retail Analysis Sam...

Sales and Marketin...

Reports

Retail Analysis Sam...

Sales and Marketin...

Workbooks

Get data

This is just the beginning!

Summary

Summary

Training A.I. Models

Building A.I. Apps

Using A.I. Tools